Mini Project - Urban Air Quality Analytics (Python)

Code:

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Pollutant Levels by City (Grouped Bar Chart)
pol_df = pd.read_csv("pollutant_data.csv")
pol_df = pol_df.set_index('City')

fig, axes = plt.subplots(3, 2, figsize=(15, 18))
plt.subplots_adjust(hspace=0.4)

pol_df.plot(kind='bar', ax=axes[0, 0], colormap='Set2')
axes[0, 0].set_title('Pollutant Levels by City')
axes[0, 0].set_ylabel('Concentration (ug/m3)')
axes[0, 0].tick_params(axis='x', rotation=45)

# 2. Smog Alert Frequency by Month
alert_freq = pd.read_csv("alert_freq.csv")
axes[0, 1].plot(alert_freq['Month'], alert_freq['Number of Alert Days'],
                marker='o', color='darkgreen')
axes[0, 1].set_title('Smog Alert Frequency by Month')
axes[0, 1].set_xlabel('Month')
axes[0, 1].set_ylabel('Number of Alert Days')
axes[0, 1].set_xticks(range(1, 13))

# 3. AQI Score Distribution by Season
aqi = pd.read_csv("aqi_scores.csv")
sns.boxplot(x='Season', y='AQI Score', data=aqi, notch=True,
            ax=axes[1, 0], palette='Set3')
axes[1, 0].set_title('AQI Score Distribution by Season')

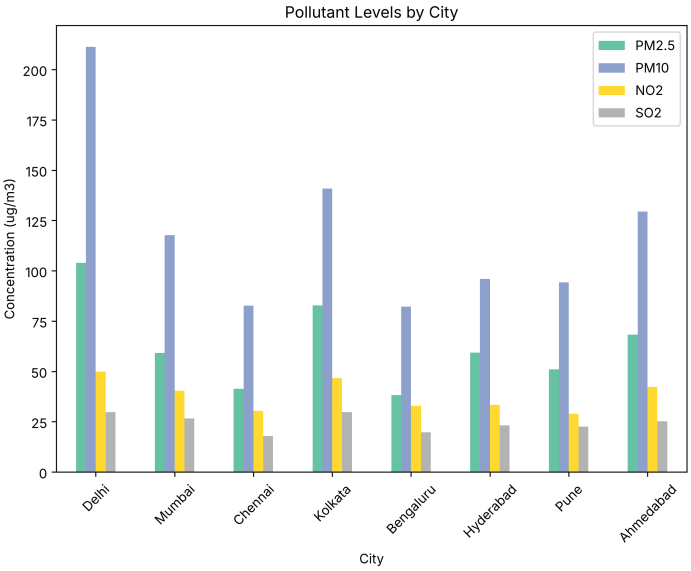
# 4. Seasonal Pollutant Drift Heatmap
drift_df = pd.read_csv("drift_data.csv")
pivot = drift_df.pivot_table(index='Origin Season',
                             columns='Destination Season',
                             values='Drift Index', aggfunc='mean')
sns.heatmap(pivot, annot=True, fmt=".1f", cmap="YlOrRd", ax=axes[1, 1])
axes[1, 1].set_title('Seasonal Pollutant Drift Transitions')

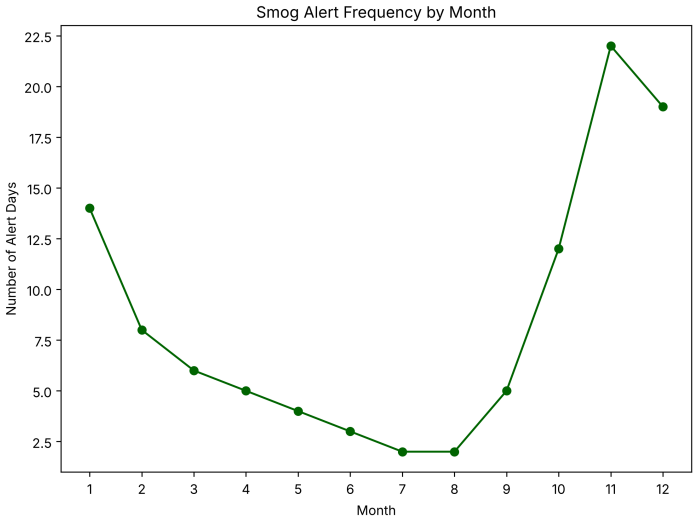
# 5. Seasonal Reading Distribution
season_counts = aqi['Season'].value_counts()
season_counts.plot(kind='bar', ax=axes[2, 0],
                   color=['lightcoral', 'navajowhite', 'lightsteelblue', 'cornflowerblue'],
                   edgecolor='black')
axes[2, 0].set_title('Seasonal Reading Distribution')
axes[2, 0].set_ylabel('Number of Readings')
axes[2, 0].tick_params(axis='x', rotation=0)
for i, v in enumerate(season_counts):
    axes[2, 0].text(i, v + 2, str(v), ha='center', fontweight='bold')

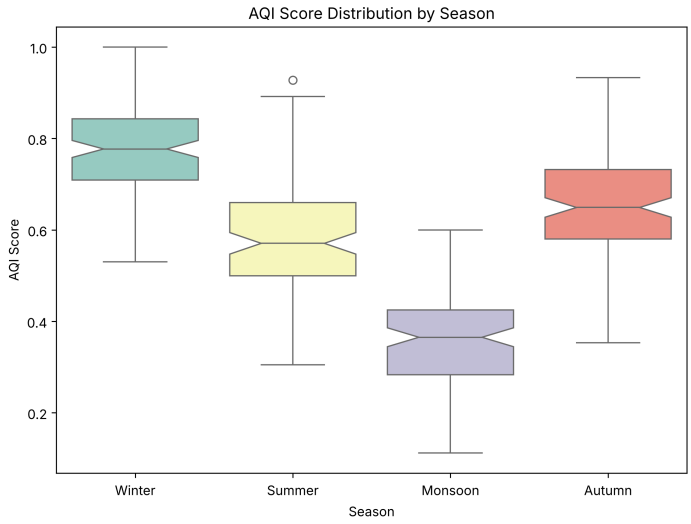
axes[2, 1].axis('off')  # Hide unused subplot

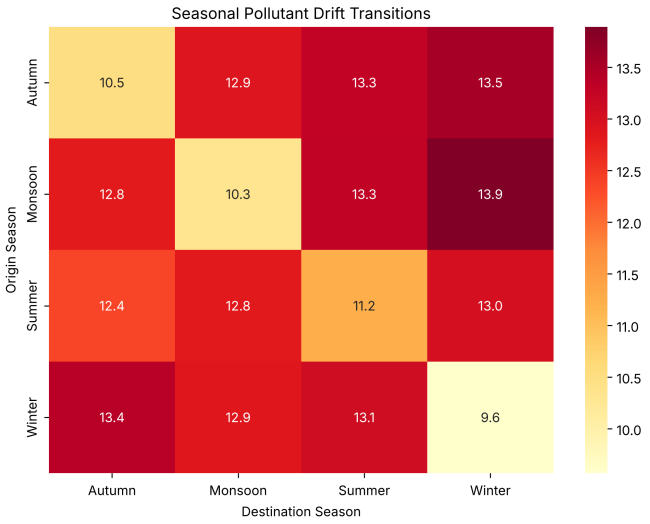
plt.tight_layout()
plt.savefig('AirQuality_Analytics_Dashboard.png', dpi=300, bbox_inches='tight')
print("Visualizations successfully saved to AirQuality_Analytics_Dashboard.png")
```

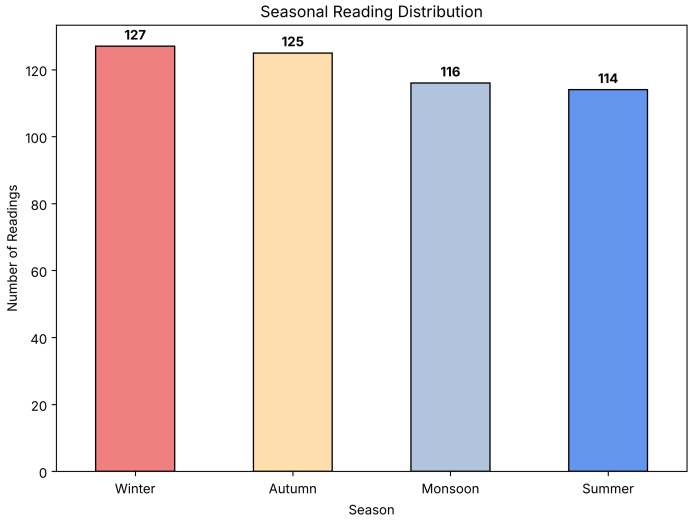
Output:











Note: the datasets used here are sample (synthetic) data generated for this exercise, not official CPCB measurements.